INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE

TESLA

SEMI 2026+

ELECTRIC



VERSION: 001

DATE: Sep 24 2026

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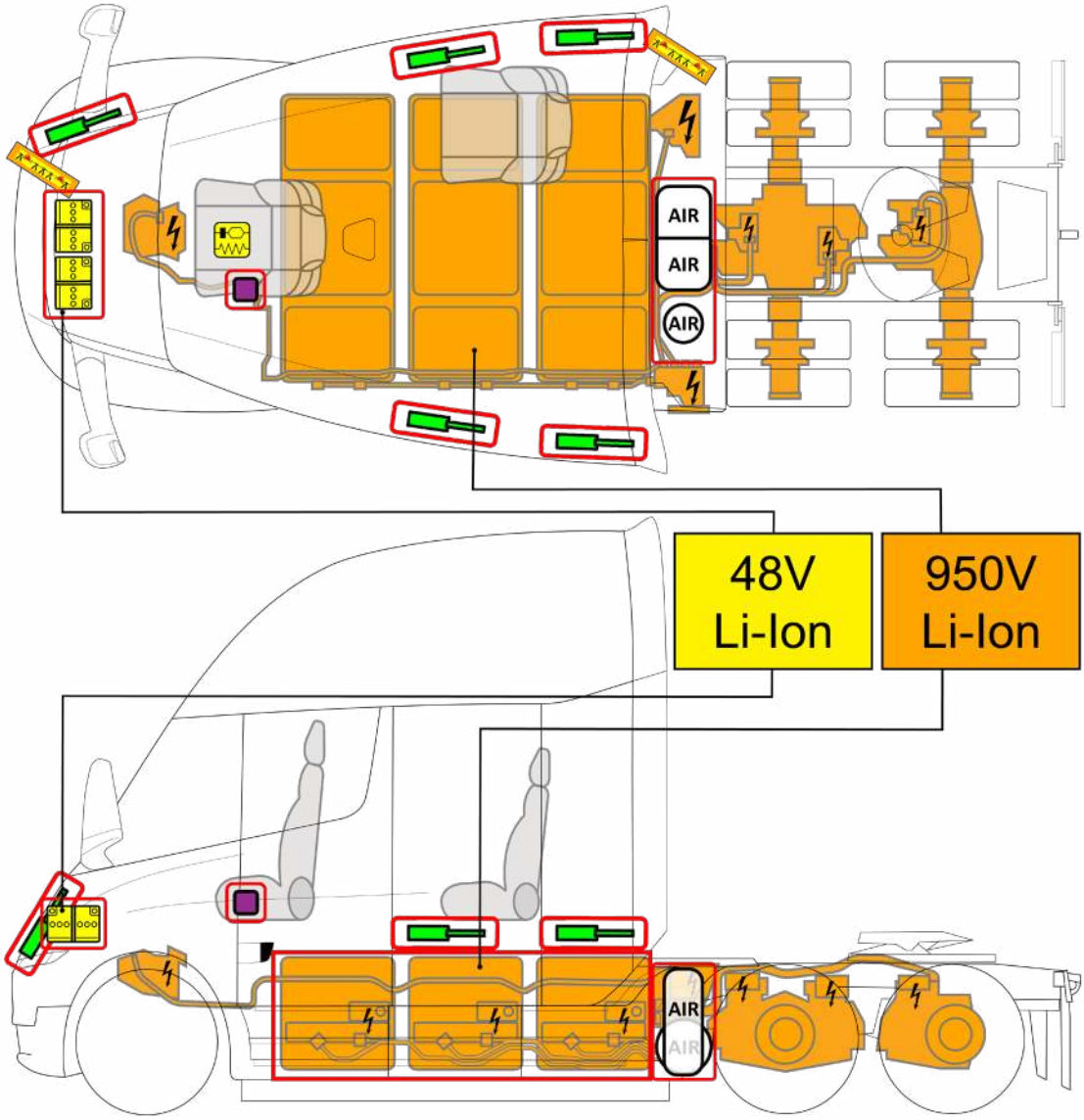
10. Explanation pictograms used Page 35

TESLA SEMI LONG RANGE

From 2026—Present

2 doors / 2 seats / semi truck





NOTE: Not all features may be present.

	High voltage component		SRS control unit	AIR	Air tank		Seat belt pre-tensioner		Gas strut / Preloaded spring
	Battery low voltage		Battery pack, high-voltage		High voltage power cable		Cable cut		

1. Identification / recognition

WARNING LACK OF ENGINE NOISE DOES NOT MEAN VEHICLE IS OFF: SILENT MOVEMENT OR INSTANT RESTART CAPABILITY EXISTS UNTIL VEHICLE IS FULLY SHUT DOWN. WEAR APPROPRIATE PPE.

Form Factor, Wheel Covers, Lightbars

Semi can be identified by the distinct lightbars, unique form factor, and side mirrors. The side mirrors are distinctively angled. The door handles are latches embedded into the door panels. There is a front grille that's smaller and lower than most trucks.

Semi is also equipped with a suite of camera sensors, with three on each side mirror, a camera above the grille, two cameras in the windshield, and an interior-facing cabin camera.

The 2026+ variant of Tesla Semi can be distinguished from older versions by the lightbars and side repeaters.







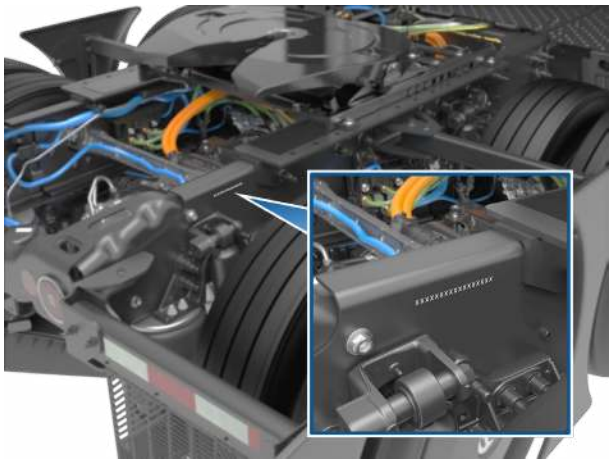
Vehicle Certification Label

The vehicle certification label, which includes the VIN, can be found on the left side door pillar.



Vehicle Identification Number (VIN)

Semi can be identified by its VIN. Open the door and locate the vehicle certification label fixed to the A-pillar. The VIN is also stamped on the side of the trailer rails at the rear of the vehicle, behind the fifth wheel.



Cabin

Semi has a unique cabin layout. The driver seat is in the center of the cabin, with touchscreens on each side in a “landscape” orientation.

A foldable passenger chair is located on the rear right of the cabin. A fire extinguisher and emergency escape ladder are fixed to the rear cabin wall to the left. There is no glovebox.



Touchscreens

Semi is equipped with two touchscreens on either side of the driver's seat. The left touchscreen features driver controls, ergonomics, and climate controls. The right touchscreen displays the navigation and configuration menu. Underneath the right touchscreen is the hazard switch, parking brake, and trailer air supply.

1. Left touchscreen for drive controls.
2. Right touchscreen for navigation and configuration.



After a collision, low voltage power may not be available and the touchscreen may not be operational. Trying to support low voltage power on a vehicle that has been in a collision or incident could lead to a possible electrical fire. Tesla does not recommend attempting to reconnect low voltage power after an accident.

Charge Port

The charge port for Semi is situated on the left side of the vehicle, to the rear, beneath the catwalk. Lightly press the charge port cap to open.

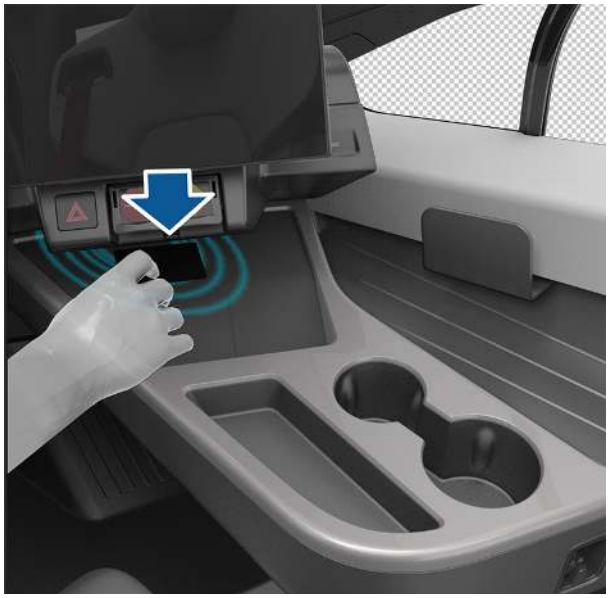
If Semi is charging and the charger cannot be removed, see "Disabling a Charging Vehicle" in Chapter 3: Disable direct hazards / safety regulations.

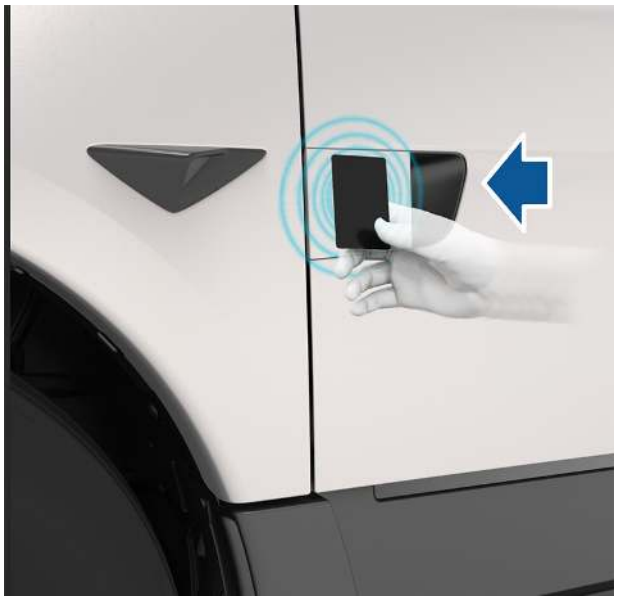


Keys

Tesla Semi supports these types of keys:

- **Authenticated phone –** A personal smartphone can be set up to communicate with Semi using Bluetooth. If a smartphone is already paired to the vehicle, open the Tesla mobile app on the smartphone and navigate to CONTROLS to unlock or lock Semi.
- **Key card –** Tap the Semi key card on any door latch to access the locked vehicle. You can also tap the key on the wireless phone charger to enable driving functionality.





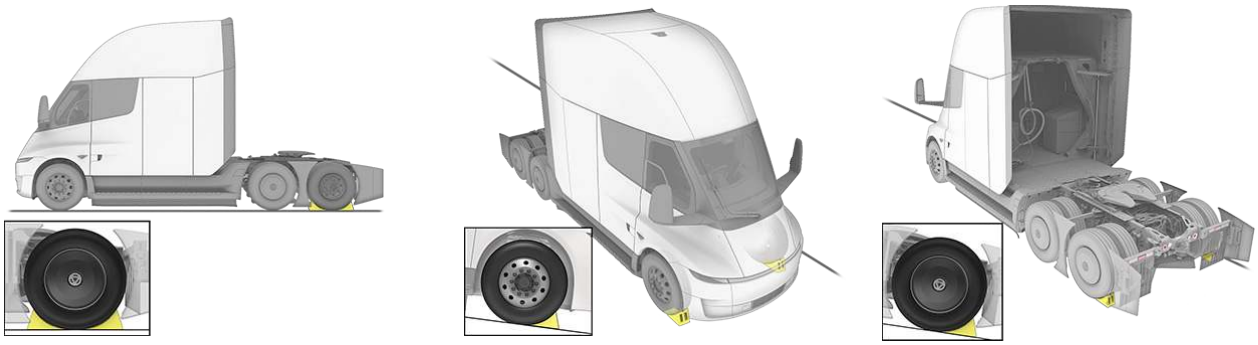
2. Immobilization / stabilization / lifting

IMMOBILIZATION

1. Chock wheels

Semi moves relatively quietly, so never assume it is powered off. Never assume that Semi will not move. Always chock the wheels. Wheels must be chocked differently depending on whether Semi is going uphill, downhill, or on level ground.

1. Level ground
2. Downhill
3. Uphill



2. Put the vehicle into Park

Pressing the accelerator pedal even slightly can cause Semi to accelerate quickly if the active gear is Drive or Reverse. To ensure that the parking brake is engaged, press and hold the Park (P) button on the touch-screen to shift into Park and enable the parking brake. You can also manually pull the tractor and trailer parking brakes situated beneath the right touchscreen. If the LED lights on each side of the brakes are solid red, the parking brakes are engaged. If they are flashing, Semi is experiencing a fault. Whenever Semi is in Park, the parking brake is automatically engaged and the touchscreen shows the active gear as Park (P).

The trailer and tractor parking brakes.



Fault detected on the tractor parking brake.



NOTE Cut or disconnect the First Responder Loop if necessary to disable the vehicle.

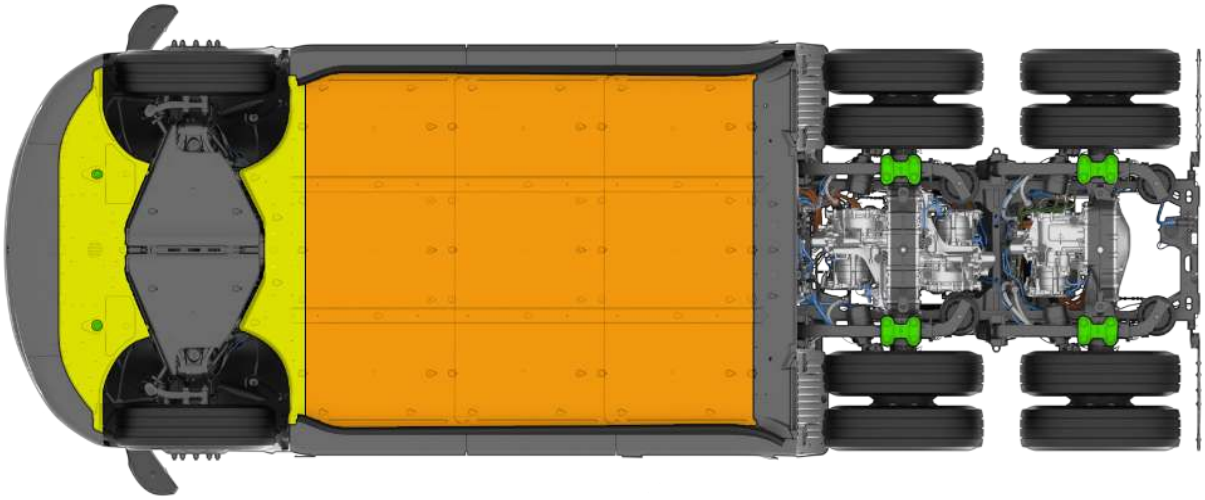
STABILIZATION / LIFTING POINTS

The high voltage battery packs are located under the vehicle cabin and catwalk. A large section of the undercarriage houses the high voltage batteries. When lifting or stabilizing Semi, only use the designated lift areas and jacking points. You can use the tow eyes stowed in the rear compartment to help lift the front. If air pressure in the suspension is lost and Semi was in a front crash, you can lift from the tires (e.g. column lifts).

WARNING Be careful to not damage the battery packs while stabilizing / lifting the vehicle.

WARNING The vehicle should be lifted or manipulated only if first responders are appropriately trained and are familiar with the vehicle's lifting points. Use caution to ensure you never come into contact with the high voltage batteries or other high voltage components while lifting or manipulating the vehicle.

WARNING DO NOT USE THE HIGH VOLTAGE BATTERIES TO LIFT OR STABILIZE SEMI.



	Appropriate lift areas (jacking points). Remove aeroshields before lifting.
	Safe stabilization points for a Semi resting on its side. Remove plastic covers if needed.
	High voltage batteries (do not lift or stabilize)

Uncouple a Trailer

You may need to uncouple the trailer to move or stabilize Semi. Follow these steps when possible to uncouple the trailer:

1. Apply the tractor and trailer parking brakes.
2. Chock the front and rear trailer tires to prevent movement during the uncoupling process.
3. Release the tractor parking brakes, while keeping the trailer parking brakes engaged.
4. Dump the air suspension on the trailer and wait for air to fully empty (see trailer manufacturer instructions).
5. Apply the tractor parking brakes.
6. Lower the trailer landing legs until they contact the ground.
7. Disconnect and stow the air and electric supply lines from the trailer.
8. On the left touchscreen, touch the **Latch** button to unlock and release the jaws of the fifth wheel from the trailer kingpin. Continue holding the button until the status updates to **Unlatched** and the release handle is fully extended.
   a. If Semi is unpowered or the **Latch** button on the touchscreen is unavailable, you can manually unlatch the fifth wheel. Lift the release handle up into the wide slot and then move the handle forward while pulling the handle out. Secure the handle in the open position using the notch on the fifth wheel.



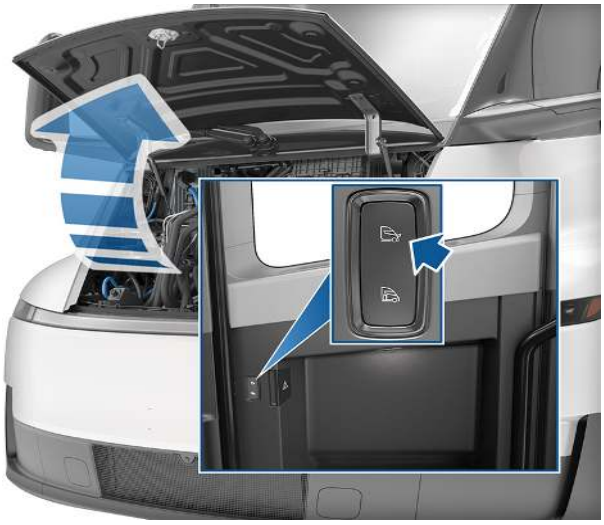
WARNING Ensure that the trailer and Semi are appropriately immobilized and stable before attempting to uncouple Semi from the trailer.

9. Once released, ensure the notch in the release handle is locked against the fifth wheel casing. From this point onwards the fifth wheel lock is held open.
10. Release the tractor parking brakes.
11. Touch **Suspension > Dump** on the left touchscreen to lower the rear axles of the tractor.
12. Drive Semi and slowly pull out from underneath the trailer.

3. Disable direct hazards / safety regulations

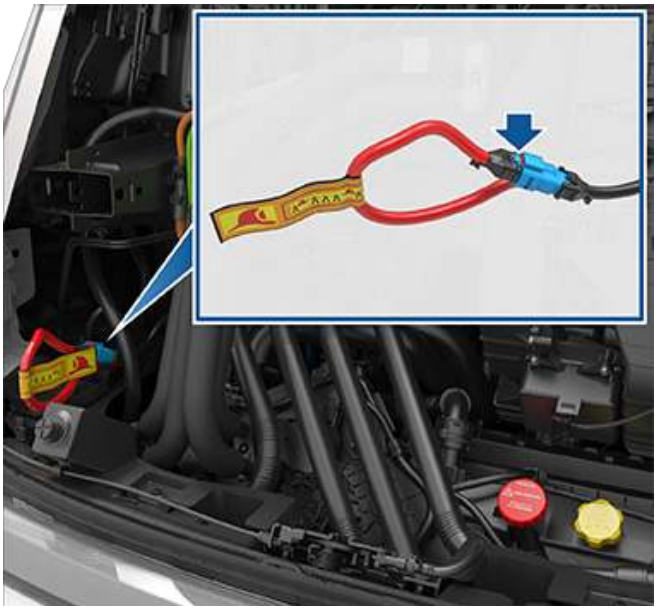
ACCESS

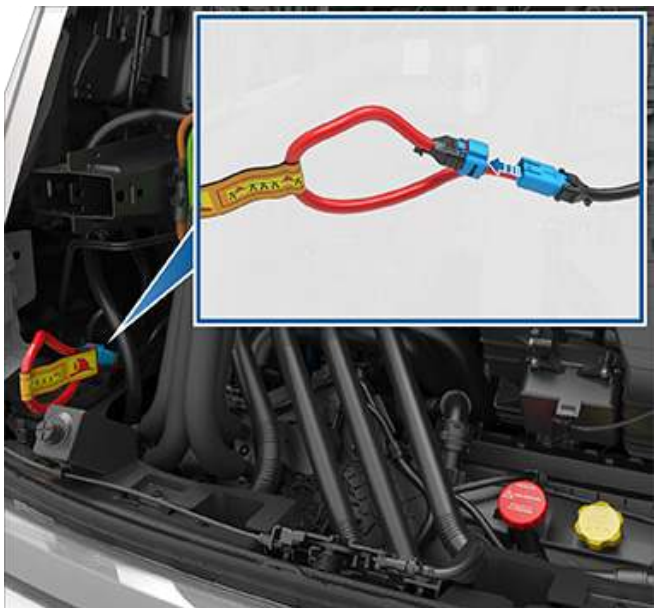
1. Open the hood (see Chapter 4: Access to the Occupants).



MAIN DISABLING METHOD

1. Always disable the first responder loop.
   a. Slide the red tab to the unlocked position and while pressing the blue tab, disconnect the loop at the connector. If you cannot pull the loop, double cut the First Responder Loop and then remove the cut section.





WARNING Not every high voltage component is labeled. Always wear appropriate PPE. If you need to cut the first responder loop, **always double cut**. Do not attempt to open the High Voltage (HV) batteries.

ALTERNATIVE DISABLING METHOD

The secondary First Responder Loop is situated beneath the catwalk on the rear right side of Tesla Semi. Remove the plastic liner in front of the middle right wheel to expose the secondary first responder loop. Then, double cut the First Responder Loop and remove the cut section.



COMPLETE DE-POWER

A complete de-power of the vehicle cuts power to the low voltage system and high voltage systems. A de-power disables all electrical systems, as well as basic vehicle functionality such as moving seats and interacting with the touchscreen. Note also that cutting the cables of the low voltage battery alone doesn't necessarily disable the low voltage system due to system redundancies. You must cut or disconnect the First Responder Loop as well to ensure that the vehicle is de-powered.

1. Shift the vehicle into Park.
2. Access the First Responder Loop under the hood or beneath the catwalk on the rear right side of the vehicle.
3. Disconnect the First Responder Loop, or double cut the loop and then remove the cut section.
4. Double cut the negative cable to **both** the low voltage batteries (black with yellow stripe).
5. Wait at least 2 minutes for the vehicle to finish de-powering.

WARNING Regardless of the disabling procedure you use, ALWAYS ASSUME THAT ALL HIGH VOLTAGE COMPONENTS ARE ENERGIZED! Cutting, crushing, or touching high voltage components can result in serious injury or death.

WARNING When using the high voltage shut down methods recommended by this document, high voltage power is isolated to the battery. The high voltage battery is always energized.

WARNING If the active gear of the vehicle is in Drive or Reverse, cutting both the low voltage battery and first responder loop might not de-power the low voltage system, due to system redundancies. High voltage power is still isolated to the battery.

First Responder Loop Cable Cut

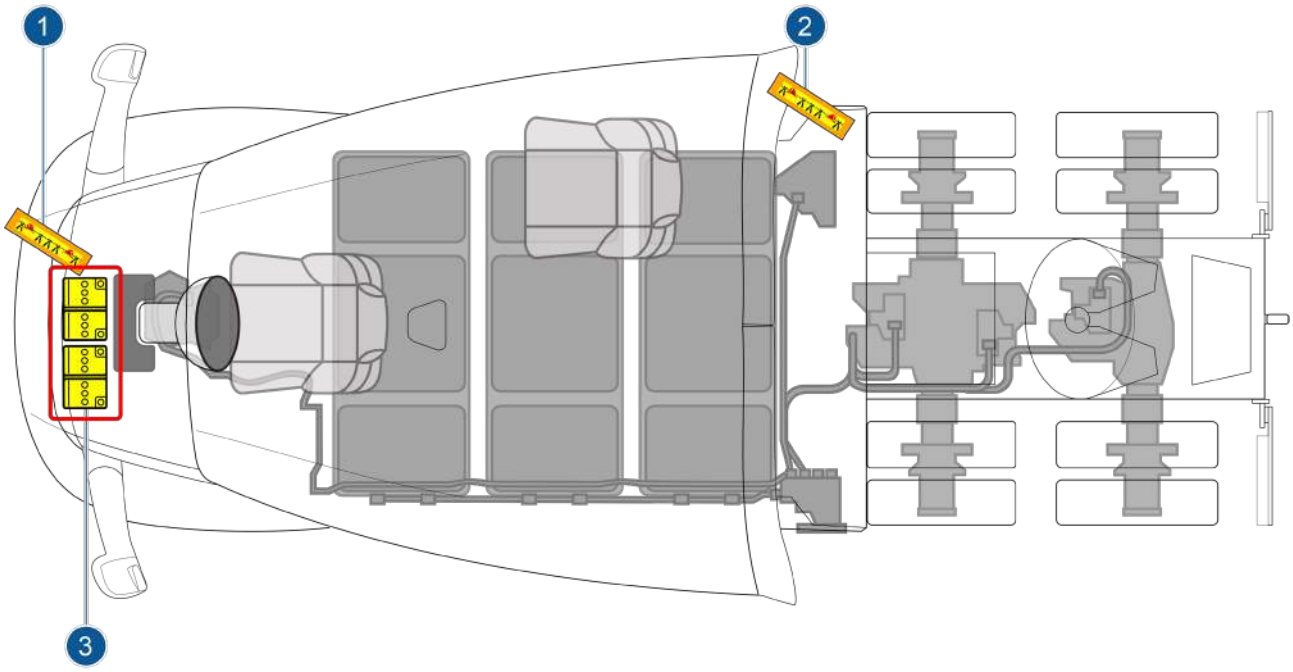
When cut or removed, the First Responder Loop removes low voltage power going to the high voltage contactors inside the high voltage battery packs, setting the high voltage contactors to “off,” or “open.” Cutting the low voltage battery cable may not disable all of the low voltage system, and it could disable the use of vehicle immobilization controls, touchscreen, and other features.

The primary First Responder Loop is located under the hood on the right side of the vehicle. There is a secondary First Responder Loop beneath the catwalk on the rear right side of the vehicle.

The high voltage contactors disconnect the High Voltage Batteries from the rest of the high voltage components, like a light switch. When “open” or in the “off” position, high voltage is present only in the battery packs. When “closed” or in the “on” position, battery pack voltage is connected to the other high voltage components. On Semi vehicles, those high voltage components include the drive units, the air conditioning compressor, the air compressor, and the charge port. When you disable the First Responder Loop, the high voltage contactors open to isolate the high voltage to the battery packs.

When the vehicle has been in an accident and the First Responder Loop has been cut or removed, always treat the packs and the high voltage components as if they are live, because the packs will still have stored energy within the cells and it is not known if other high voltage components have been damaged. Treat every orange cable and battery pack as if there is high voltage in them. Never cut an orange high voltage cable or cut into the battery pack.

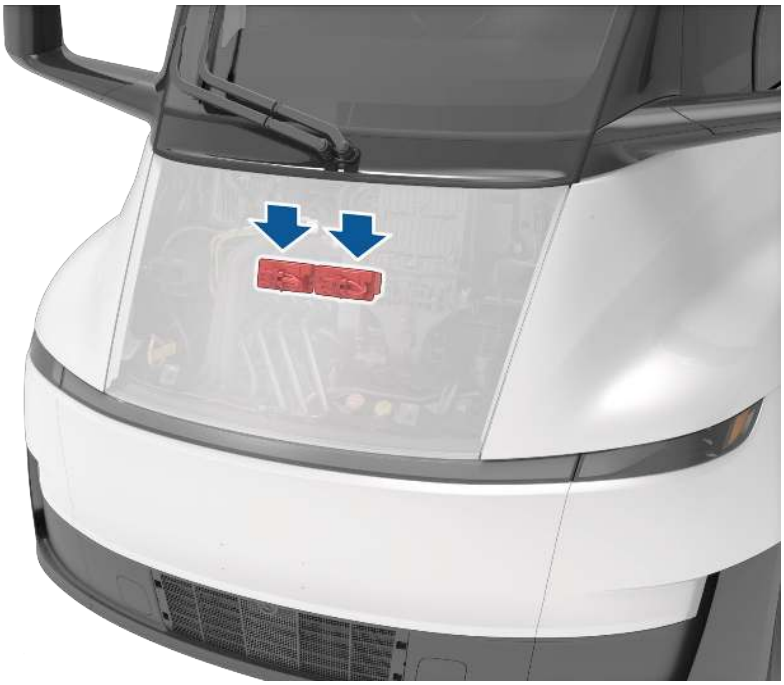
There is no way to instantaneously discharge the energy that is inside of the battery pack when a vehicle is in an accident. There is stored energy in battery cells. Caution must be used to not damage the battery packs in the case of vehicle extrication operations.



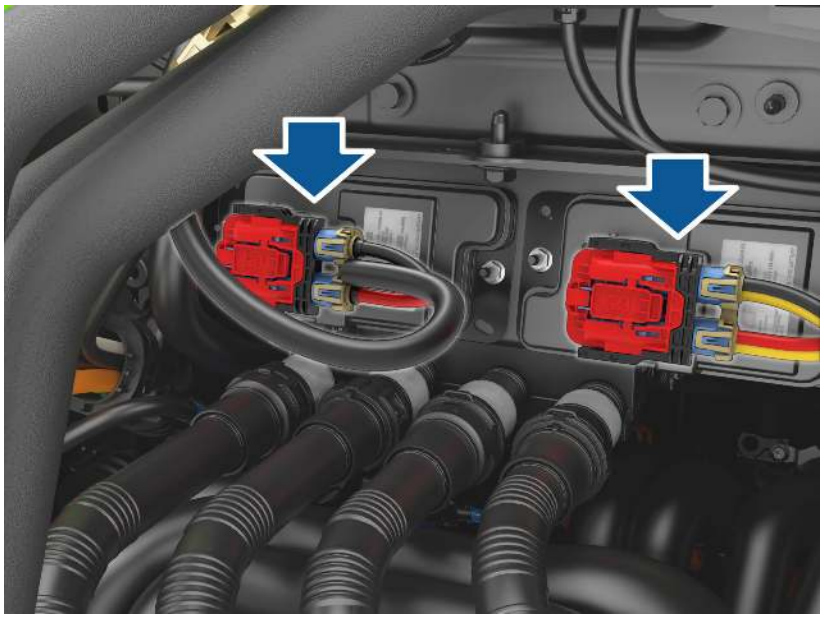
1. Primary First Responder Cut Loop
2. Secondary First Responder Cut Loop
3. 48V Low Voltage Battery

Access to Low Voltage Batteries

Open the vehicle hood to access the two 48V low voltage batteries. When necessary, both the negative battery cables (black with a yellow stripe) should be double cut to open the low voltage battery circuit. Care should always be taken to not cut both the positive and negative battery cables at the same time when double cutting the negative battery cable. Cutting both cables simultaneously can short circuit the low voltage battery. Wait at least 60 seconds for the low voltage system to discharge.



The low voltage batteries. Note the black negative cable with a yellow stripe.



WARNING Cutting the low voltage battery cables doesn't always disable the low voltage system. Semi is designed with redundancies that use the high voltage system to power the low voltage system.

Disabling a Charging Vehicle

Before attempting to disable the vehicle, unplug the charging cable. If you have access to the cabin, touch **Controls > Charging > Unlock Charge port** on the right touchscreen. You can also hold down the button on the charging cable of Tesla chargers if the vehicle is unlocked or a paired key is nearby to unplug the cable without accessing the vehicle cabin.

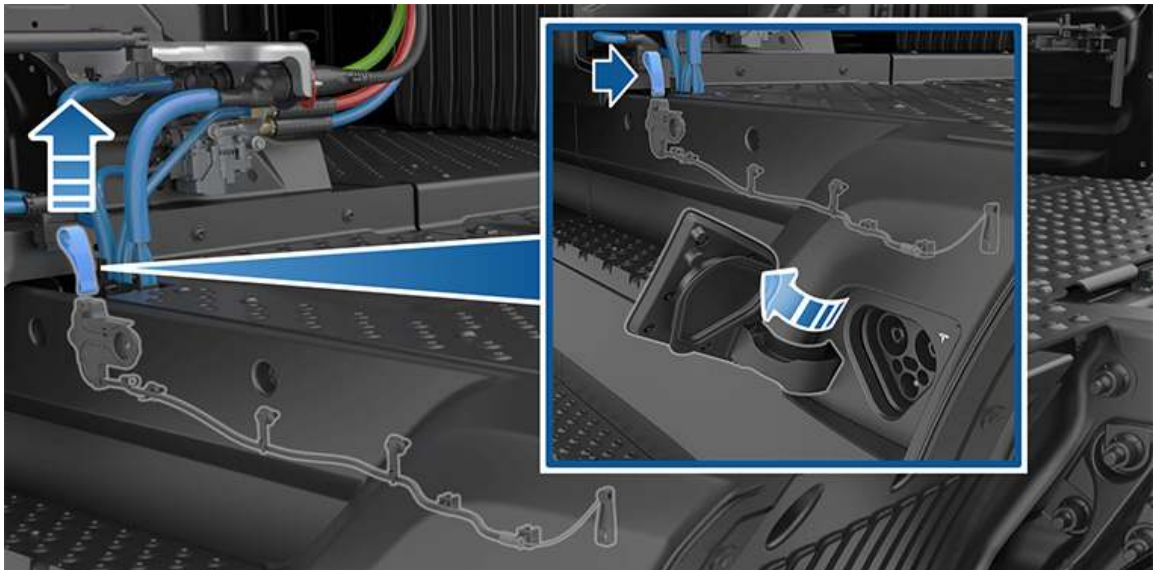
If the vehicle cannot be unplugged due to a damaged or locked charging cable, turn off power to the charging station. Then, proceed with disabling procedures by double cutting the First Responder Loop.

WARNING Do not cut the charging cable while the charging station has power. Cutting the charging cable while there is still power can start an electrical fire or cause severe injury.

Manually Release Charging Cable

If you cannot unplug the charging cable and cannot disable the charging station, you can attempt to manually release the charging cable instead of cutting it.

1. On the left side of the vehicle, pull the strap in the catwalk straight up.

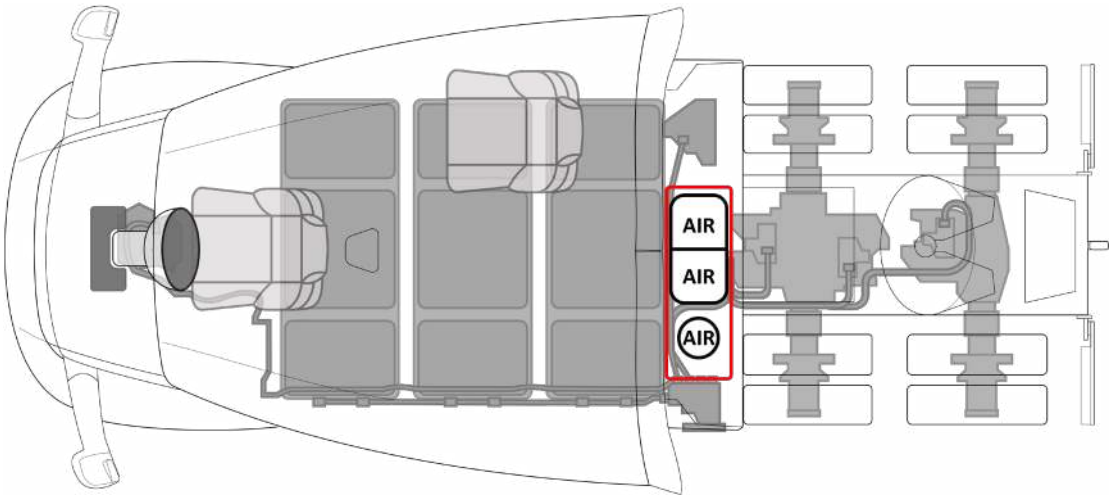


2. Pull the charging cable from the charge port.

WARNING Do not pull the manual release cable while simultaneously attempting to remove the charge cable from the charge port. Always pull the release cable before attempting to remove the charge cable. Failure to follow these instructions can result in electric shock and serious injury.

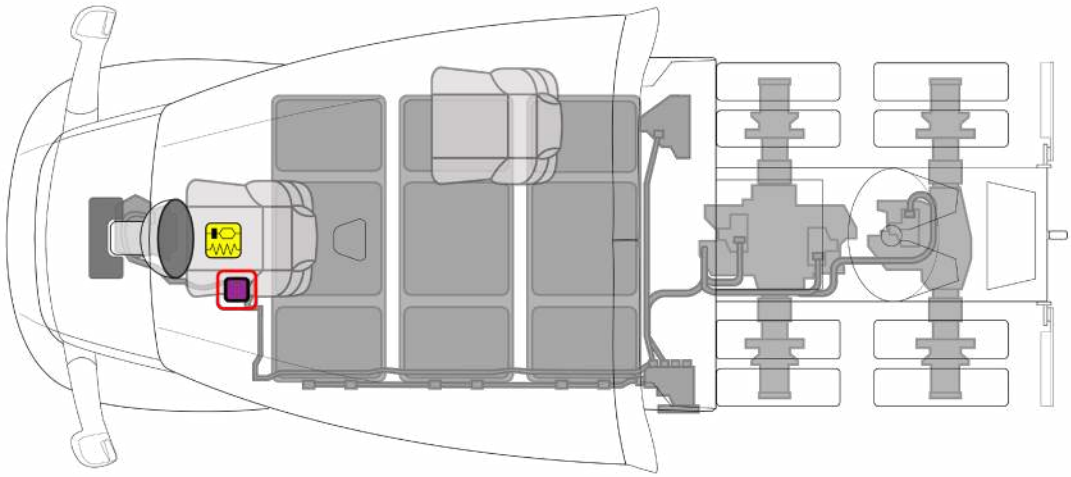
AIR Air Tanks

There is a dual volume air tank located beneath the catwalk. There is a second vertical air tank beside the left frame rail beneath the catwalk. The air tanks can be manually drained with three release cables located on the left side of the vehicle, below the side panel and beneath the catwalk.



Seat Belt Pre-Tensioners

The seat belt pre-tensioner, outlined in red, is located at the bottom of the driver's seat. The Restraint Control Module (RCM) is located beneath the driver's seat. In addition to the traditional seat belt pre-tensioner, Semi has a motorized seat belt that can automatically tighten to reduce slack when buckled or in anticipation of a crash. The passenger seat is not equipped with a seat belt pre-tensioner.



WARNING The RCM has a backup power supply with a discharge time of approximately 10 seconds. Do not touch the RCM within 10 seconds of a pre-tensioner deployment.

WARNING Electrical and mechanical releases may be compromised after a collision.

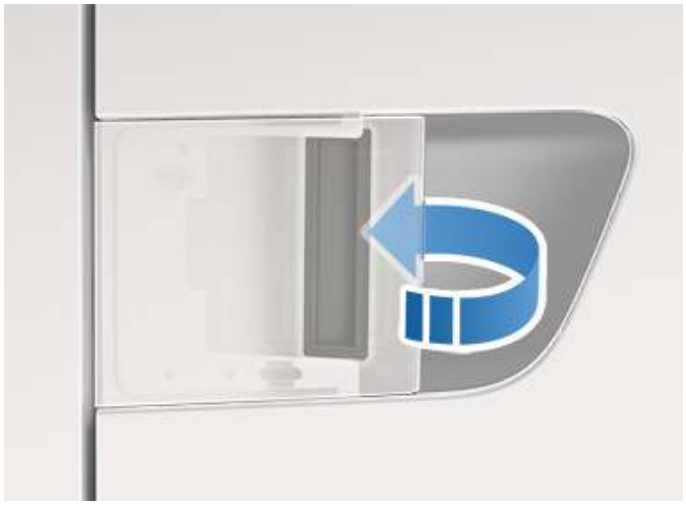
4. Access to the occupants

NOTE: The seats and touchscreen are electrically powered and may not function after a collision.

NOTE: After a collision, the doors are designed to automatically unlock. If there is insufficient low voltage power, there is a chance the doors do not unlock as intended. Severe collisions may cause doors to be mechanically difficult to operate. Extrication may be required.

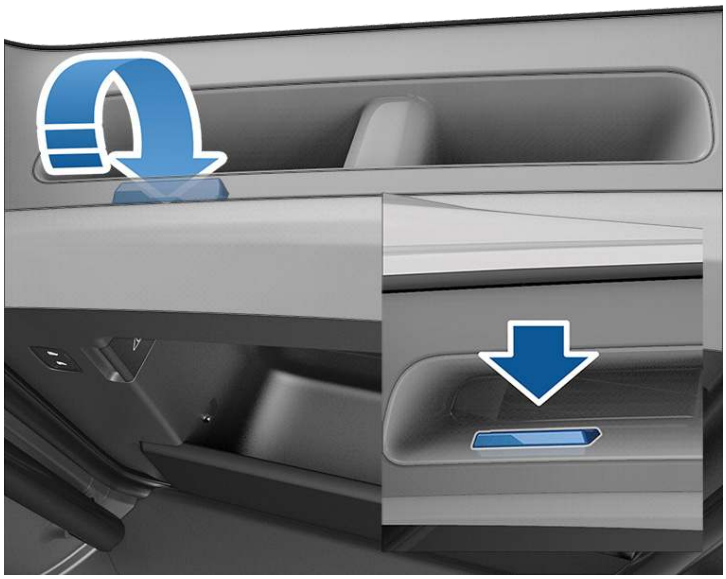
Opening Doors from the Outside with Power

To open the Semi doors from the outside, press the door handle switch and then pull the door open.



Opening Doors from Inside with Power

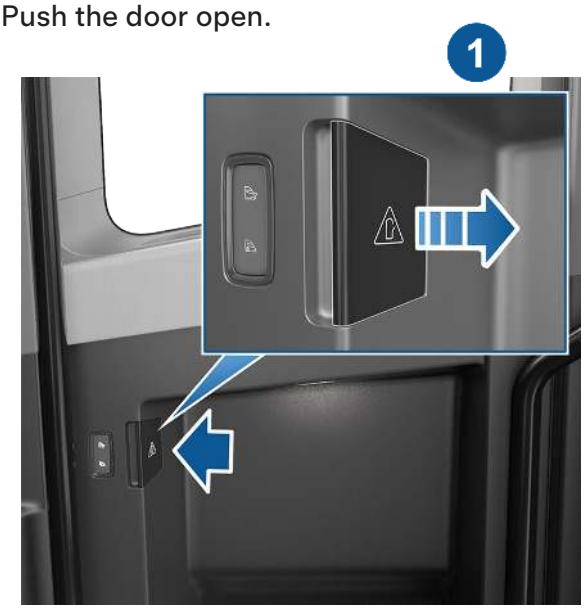
To open Semi doors from inside with low voltage power available, lift the latch located near the door panel.

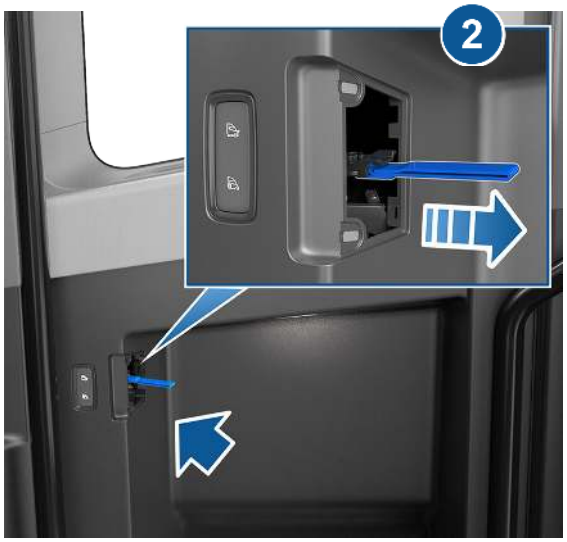


Opening Doors from Inside without Power

To open the Semi doors from the inside without low voltage power:

1. Pull off the cover of the mechanical release.
2. Pull on the strap to engage the mechanical release.
3. Push the door open.

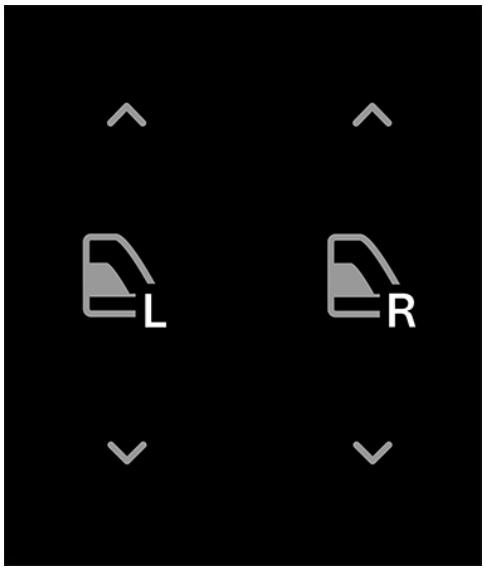




It is important to know also that in any vehicle collision with damage to the front doors, the mechanical door release may not operate as designed. Remember also that every vehicle accident is different and may require extrication operations to gain access to the vehicle's cabin.

Rolling Down Windows with Power

The door windows are fixed and cannot be rolled down. There is a small glass panel in the front quarter glass pane on each side called venting windows that can roll down via the left touchscreen. Venting windows are too small for extrication or entry and are typically used for toll booths, key cards, or badge access to facilities, etc. The venting windows can only move when low voltage power is enabled, and can still be used after the First Responder Loop is cut or disconnected if the low voltage system is intact.



Moving the Seats with Power

Semi has electrically powered seats that move with buttons located on the side of the seat. The buttons operate only when low voltage power is enabled and can still be used after the First Responder Loop is cut or removed if the low voltage system is intact. Vertical adjustment of seats uses air, while horizontal movement is motorized.

1. Moves seat forward/backward and adjust the height and tilt angle of the seat up/down.
2. Adjusts backrest.
3. Adjusts lumbar support.



Emergency Escape Ladder

Semi is equipped with an emergency escape ladder to help occupants exit the vehicle in situations where Semi has rolled. The escape ladder is typically fixed to the rear wall of the cabin. The escape ladder has a maximum weight capacity of 450 lbs and is equipped with a glass breaker at the tip of the left handle.

If you need to break the door glass:

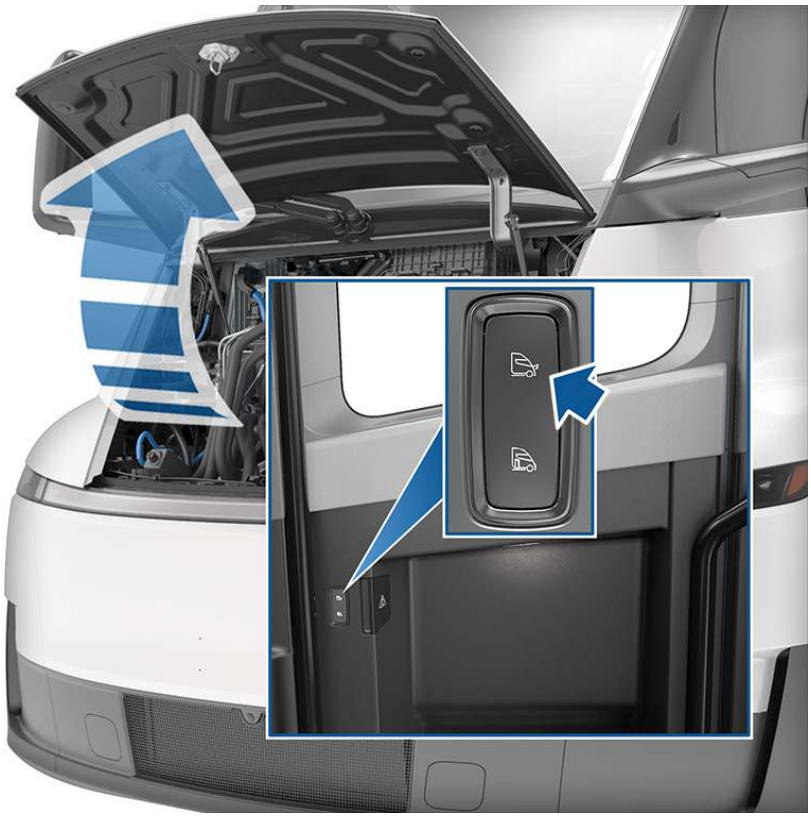
1. Unbuckle the release strap mechanism and pull the ladder free.
2. Remove the tamper sticker and plastic cap from the glass breaker.
3. Strike the door glass in a direction away from yourself.
4. After breaking the glass, clear the opening of any broken glass.
5. Extend the ladder fully and ensure the rungs lock.
6. Guide the ladder through the opening and ascend. Brace the ladder with both hands as you move through the opening.



Opening the Hood with Power

Semi does not have a traditional internal combustion engine. The space instead houses the low voltage batteries, thermal compressor, first responder loop, coolant pipelines, and various other high voltage cabling.

To open the hood with low voltage power available, press the top button on the panel inside of either door in Semi, or touch **Hood** on the left touchscreen.

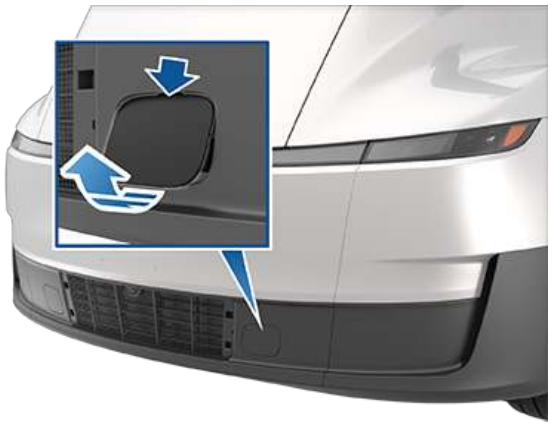


Jump Starting or Opening the Hood with a 12V Low Voltage Power Supply

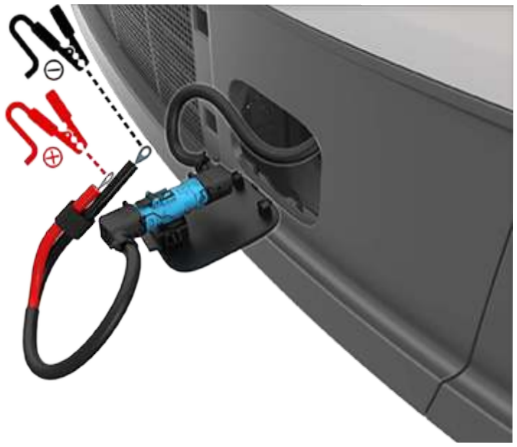
If low voltage power is not available, you might need an external power supply to access the hood and disable the first responder loop. This method will not open the hood if Semi is locked and has low voltage power. Use an external power supply with at least 12V output, such as a jumper box, for the following:

NOTE: Jumping Semi with at least 12V external power supply opens the hood and enables you to unlock and open the right passenger side door of Semi. Jumping with 12V does NOT power the touchscreen.

1. Remove the cover to the jump connector.



2. Extract the jump connector cables.
3. Connect the external power supply to the jump connector cables.
4. Turn on the external power supply (refer to the manufacturer's instructions of your external power supply). The hood latch immediately releases and you can open the hood.



5. Turn OFF the external power supply.
6. Disconnect the external power supply from the jump connector, beginning with the black negative (-) cable.

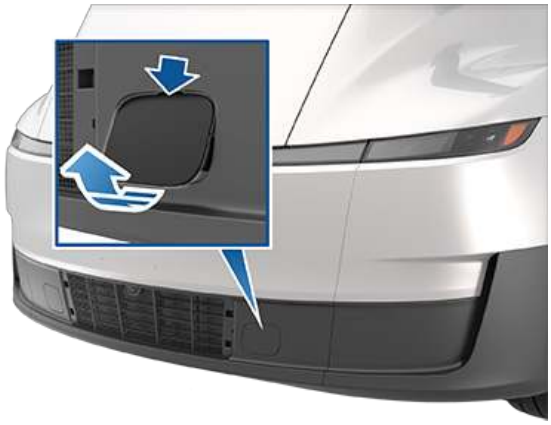
WARNING Connecting or disconnecting the jump start connector while the external power supply is ON can cause arcing. Approach cautiously. Wear appropriate PPE.

Jump Starting or Opening the Hood with a 48V Low Voltage Power Supply

If low voltage power is not available, you might need an external power supply to jump Semi and/or access the hood to disable the first responder loop. This method will not open the hood if Semi is locked and has low voltage power. Use an external power supply with 48V output, such as a jumper box, for the following:

NOTE: Jumping Semi with a 48V external power supply opens the hood and powers the doors, touchscreens, and seat controls.

1. Remove the cover to the jump connector.



2. Extract the jump connector cables.
3. Remove the 12V connector toggles on the jump connector to expose the 48V connector cables.
4. Connect the external power supply to the jump connector cable and ensure the connector locks.
5. Turn on the external power supply (refer to the manufacturer's instructions of your external power supply). The hood latch immediately releases and you can open the hood.



6. Turn OFF the external power supply.
7. Disconnect the external power supply from the jump connector.

WARNING Connecting or disconnecting the jump connector while the external power supply is ON can cause arcing. Approach cautiously. Wear appropriate PPE.

8. Underneath the hood of Semi, locate the jump connector that powers the left side of the vehicle.
9. Remove the dust cap on the jump connector. Press down on lock tab and pull the connector outwards.
10. Repeat steps 4–7 for the left side and right side jump connector cables. Keep the external power supply connected for at least 2 minutes to power the touchscreen and low voltage system.



NOTE: Tesla recommends jumping both the left and right sides simultaneously with two separate 48V external power sources for a full recovery.

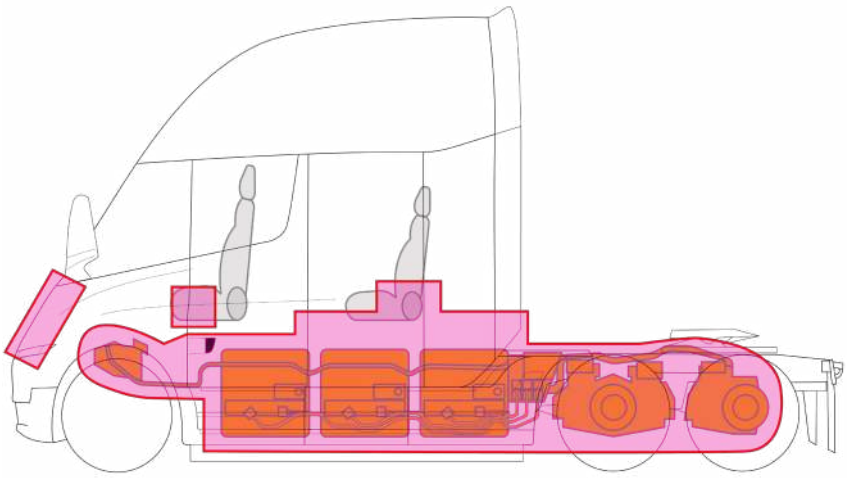
High Strength Zone

The frame of Semi is reinforced with high strength steels. Much of the cabin is made of aluminum and is not reinforced with high strength steels. Suitable tools must be used to cut or crush the aluminum to access the cabin. Cutting of the frame is typically unnecessary and is not recommended, as it can compromise the high voltage battery or damage the high voltage cabling and components.

WARNING Regardless of the disabling procedure you use, ALWAYS ASSUME THAT ALL HIGH VOLTAGE COMPONENTS ARE ENERGIZED! Cutting, crushing, or touching high voltage components can result in serious injury or death.

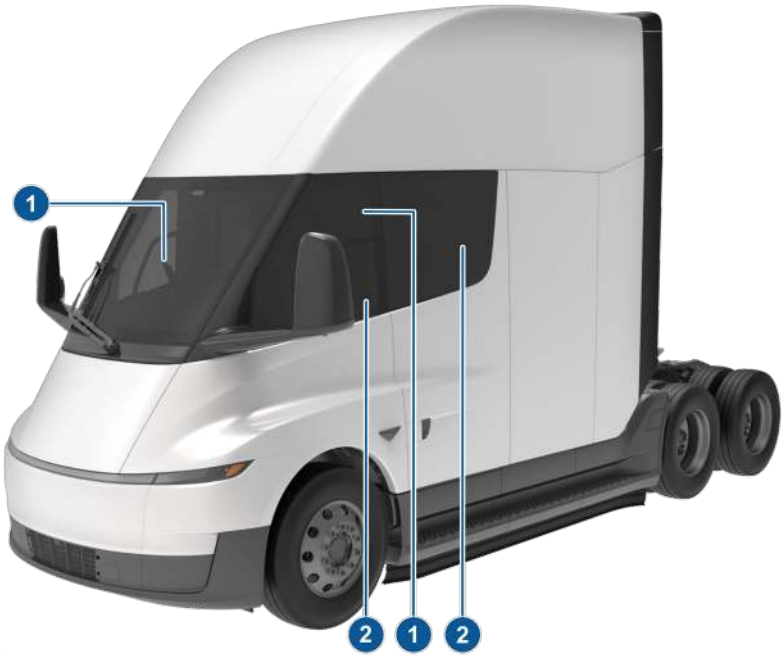
No-Cut Zones

Semi has areas that are defined as “no-cut zones” due to the presence of high voltage, gas struts, SRS components, or other hazards. Never cut or crush in these areas. Doing so could result in serious injury or death. The “no-cut zones” are shown in pink.



Windows

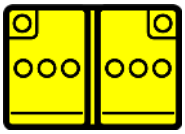
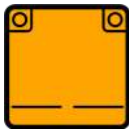
The windshields on Semi are made of laminated safety glass. The venting windows and fixed door glass are tempered safety glass. The glass breaker on the escape ladder is designed to break the fixed door glass.



1. Laminated safety glass
2. Tempered safety glass

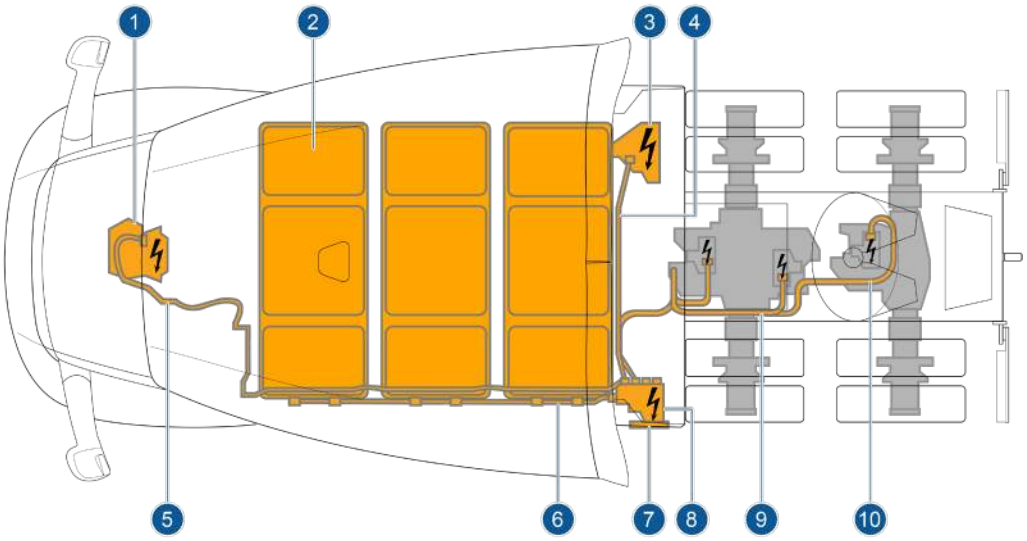
NOTE: Cutting laminated safety glass can create a lot of glass dust. Consider additional respiratory PPE for both crew and passengers when cutting laminated glass.

5. Stored energy / liquids / gases / solids

		48V
		950V

WARNING Clear liquid is likely water. Battery electrolyte is clear, but battery cells are sealed and have a limited volume of electrolyte per cell. The coolant is orange.

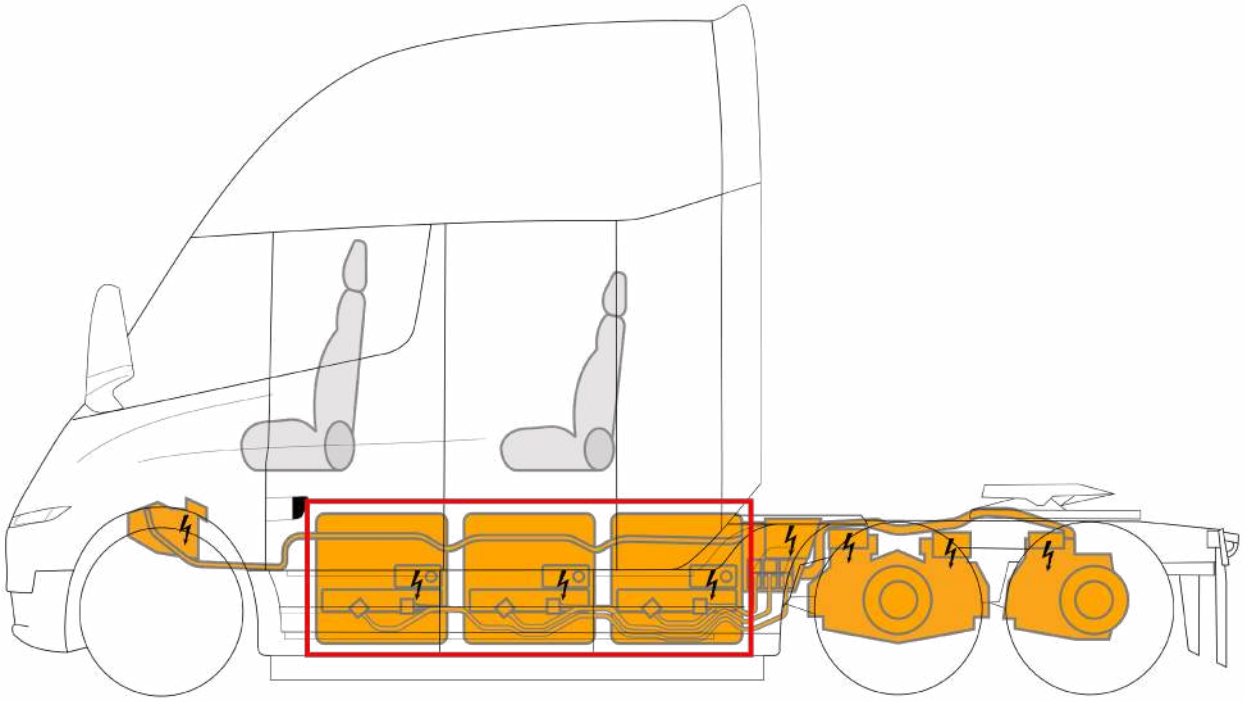
High Voltage Components



1. Thermal Compressor
2. High Voltage Battery Packs (depends on configuration)
3. Air Compressor
4. High Voltage Cables
5. High Voltage Cables
6. High Voltage Busbars and Cables
7. Charge Port
8. High Voltage Junction Box
9. High Voltage Cables to Drive Unit 1
10. High Voltage Cables to Drive Unit 2

High Voltage Battery Packs

Semi can be equipped with three 950V lithium-ion high voltage battery packs connected in parallel. The batteries are made up of many cells that are liquid cooled with coolant. The coolant will appear orange in color and may leak from the battery packs if the packs have been compromised during a vehicle collision. The battery cells will have stored energy within them. Never breach the high voltage batteries when lifting from under the vehicle. When using rescue tools, pay special attention to ensure that you do not breach any batteries from the vehicle cabin or catwalks. Refer to Chapter 2: Lift Areas for instructions on how to properly lift the vehicle.



Pushing on the Cabin Floor

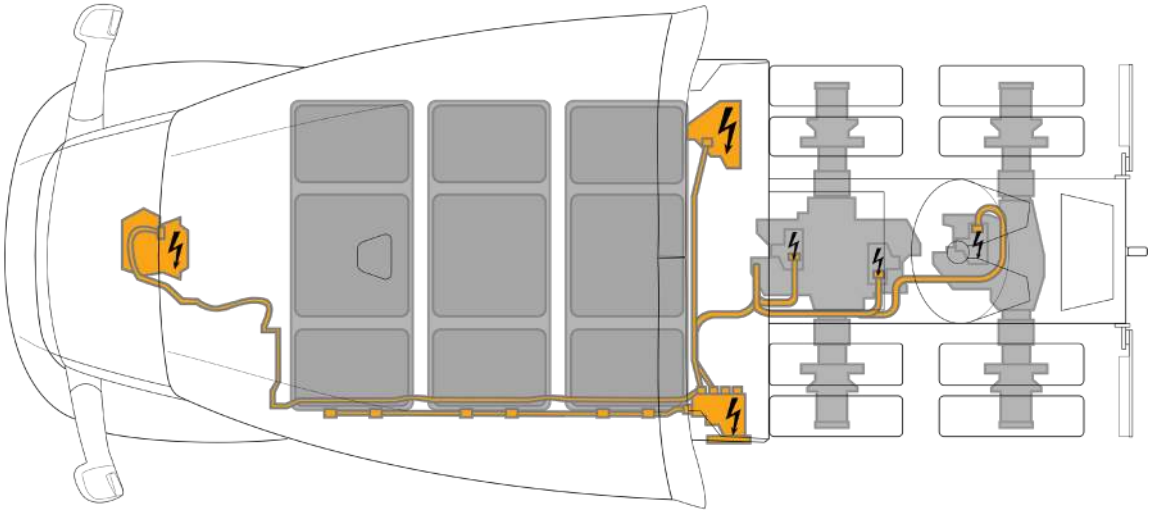
The first two high voltage batteries are located below the floor pan in the vehicle cabin. The last high voltage battery is situated beneath the exterior rear deck of the vehicle. At no time should the high voltage battery pack be compromised with rescue tools.

WARNING Never push on the floor pan from the cabin or catwalk or exert downward force with a sharp rescue tool. Doing so can breach the high voltage battery or damage the high voltage cables, which can cause serious injury or death.

High Voltage Power Cable / Busbar / Component

High voltage cables and busbars are shown in orange. There are high voltage cables and busbars that run the length of the battery packs on the left side and extend through to the drive units. The busbars might have an unfinished aluminum surface.

> **WARNING** Do not compromise these high voltage components with rescue tools. The assumption should be made that there may be high voltage present in the orange high voltage cables and unfinished busbars at all times.

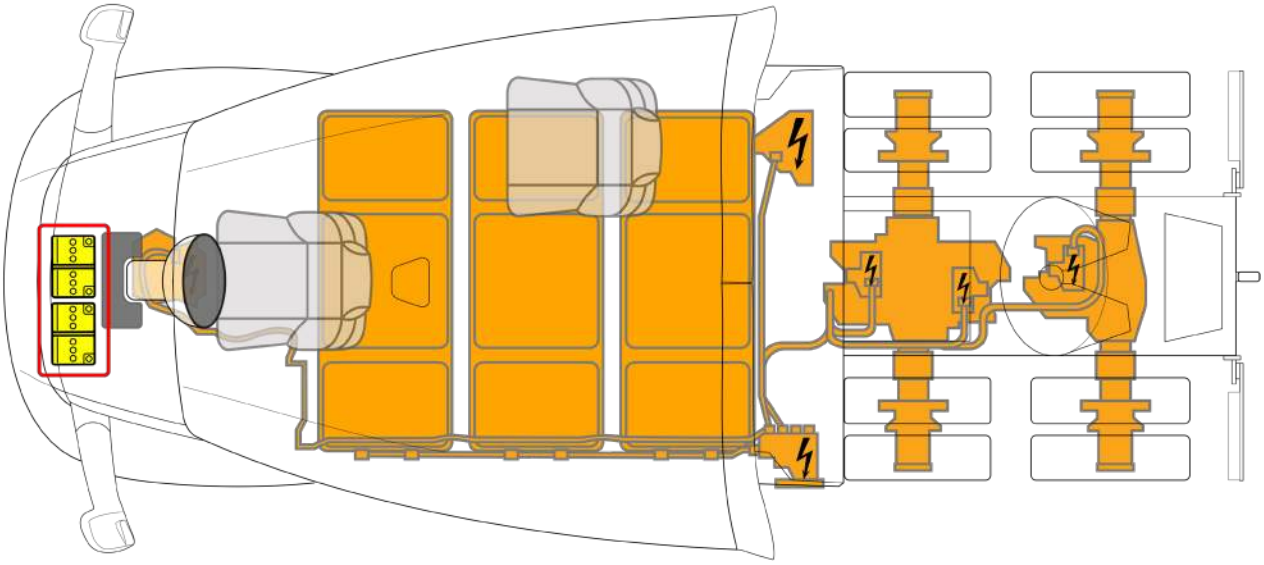


Drive Unit

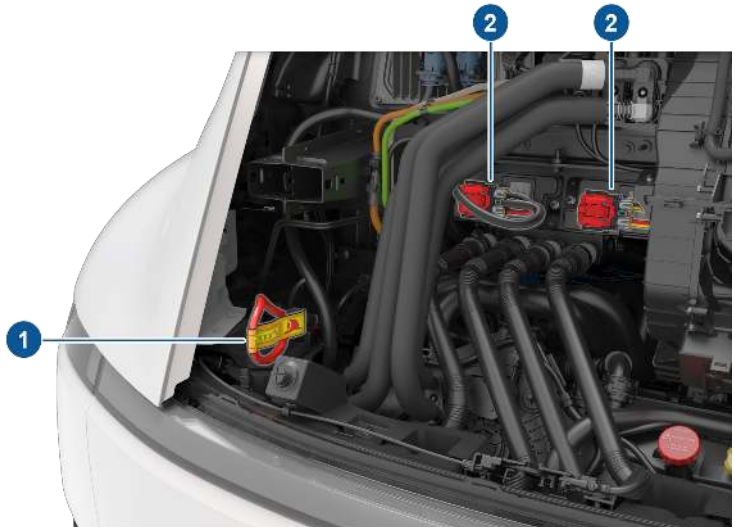
There are three drive units located to the rear, past the catwalk. These consist of left and right drive units for the middle wheels, and a drive unit for the two rear wheels. The drive inverter is located within the drive unit. The drive units convert Direct Current (DC) from the high voltage battery into Alternating Current (AC) that the drive units use to power the wheels.

Low Voltage Battery Packs

In addition to the high voltage system, Semi has a low voltage electrical system. The two batteries of the low voltage electrical system are charged by the high voltage system. The low voltage batteries operate the restraint system, seats, windows, door locks, touchscreen, and interior and exterior lights. The low voltage batteries, outlined in red, are located under the hood.



WARNING Cutting the low voltage battery cables doesn't always disable the low voltage system. Semi is designed with redundancies that use the high voltage system to power the low voltage system.



1. First Responder Loop
2. Low Voltage Batteries

6. In case of fire

Firefighting





DO NOT SUBMERGE VEHICLE TO EXTINGUISH/COOL BATTERY FIRE

USE LARGE AMOUNTS OF WATER

POSSIBLE BATTERY RE-IGNITION!

MONITOR HV BATTERY TEMPERATURE FOR AT LEAST 24 HOURS



USE WATER TO FIGHT A HIGH VOLTAGE BATTERY FIRE. If the battery catches fire, is exposed to high heat, or is generating heat or gases, use water to cool the battery. It can take more than 8,000 gallons (30,283 liters) of water, applied directly to the battery, to fully extinguish and cool down a battery fire; always establish or request additional water supply early. If water is not immediately available, use CO2, dry chemicals, or another typical fire-extinguishing agent to fight the fire until water is available.

WARNING Do not use firefighting foams. Using firefighting foams on electric vehicles has a higher risk of electrical fire.

Apply water directly to the battery. If safety permits, lift or tilt the vehicle for more direct access to the battery (see Chapter 2: Immobilization / stabilization / lifting). Water may be applied to the interior of the pack from a safe distance ONLY if a natural opening (such as a vent or opening from a collision) already exists. Do not open the battery for the purpose of cooling it.

Tesla does not recommend placing the vehicle in a large container full of water. The use of a Thermal Image Camera or Infrared (TIC or IR) is recommended to monitor battery temperatures during the cooling process. Continue to use water until the battery has reached ambient temperatures or below, indicated by the thermal image camera. When utilizing a thermal image camera, allow enough time, once the application of water has stopped, to allow for heat within the battery to transfer to the battery enclosure.

NOTE: Semi is equipped with infrared webbing on the seat belts for visibility of passengers at night or in the dark. This infrared webbing can show up as a hot spot under TIC or IR.

Extinguish fires that do not involve the high voltage battery using typical vehicle firefighting procedures.

During fire extinguishing, do not make contact with any high voltage components. Always use insulated tools for fire extinguishing.



Heat and flames can compromise airbag inflators, stored gas inflation cylinders, gas struts, and other components which can result in unexpected excessive heat, which can cause inflation cylinder explosion. Perform an adequate knock down before entering a hot zone.



Battery fires can take over 24 hours to fully cool. After suppression and smoke has visibly subsided, a thermal image camera can be used to actively measure the temperature of the high voltage battery and monitor the trend of heating or cooling. There must be no fire, smoke, audible popping/hissing, or heating present in the high voltage battery for at least 45 minutes before the vehicle can be released to second responders (such as law enforcement, vehicle transporters, etc.). The battery must be completely cooled before releasing the vehicle to second responders or otherwise leaving the incident.

There is always a risk of battery re-ignition. Drain excess water out of the vehicle by tilting or raising a side of the vehicle approximately 1 foot (30 cm). This operation can assist in mitigating possible re-ignition.

> **WARNING** When lifting and/or moving a waterlogged vehicle to drain the HV battery, stand at least 5 feet (1.52 m) away from the HV battery and wear appropriate voltage-rated PPE.

Due to potential re-ignition, a Semi that has been involved in a submersion, fire, or a collision that has compromised the high voltage battery should be stored in an open area at least 150 feet (45 m) from any other vehicles, structures, or flammable objects or materials.

> **WARNING** During all firefighting activities, consider the vehicle energized. Always wear full PPE.

High-Voltage Battery — Fire Damage



Similar to conventional and other electric and hybrid vehicles, a burning battery releases super-heated gases and toxic vapors. This release may include volatile organic compounds, hydrogen gas, carbon dioxide, carbon monoxide, soot, particulates containing oxides of nickel, aluminum, lithium, copper, cobalt, and hydrogen fluoride. Responders should always protect themselves with full PPE, including a SCBA, and take appropriate measures to protect civilians downwind from the incident.



A damaged high voltage battery can create rapid heating of the battery cells. If you notice smoke, steam, or audible popping or hissing coming from the high voltage battery, assume that it is heated and take appropriate action as described above.

7. In case of submersion

Treat a submerged Semi like any other submerged semi truck. The body of Semi does not present a greater risk of shock because it is in water. However, handle any submerged vehicle while wearing the appropriate PPE for water rescue. Remove the vehicle from the water and continue normal disabling procedures as outlined in Chapter 3: Disable direct hazards / safety regulations.

Vehicles that are waterlogged or have been submerged in water should be handled with greater caution due to the potential risk of electrical fire. After removing the vehicle from the water and completing disabling procedures, tilt or raise one side of the vehicle approximately 1 foot (30 cm) to allow water to drain out of the vehicle and the high voltage battery packs.

WARNING When lifting and/or moving a waterlogged vehicle to drain the HV battery, stand at least 5 feet (1.52 m) away from the HV battery and wear appropriate voltage-rated PPE.

Submerged or waterlogged vehicles must be transported and stored more carefully. When loading the vehicle for towing, use a Thermal Image Camera or Infrared (TIC or IR) to monitor battery temperatures and check for potential hot spots and continue monitoring until the vehicle is taken away. Store the vehicle outside at a safe distance at least 150 feet (45 m) from other vehicles, structures, or flammable objects or materials.

WARNING Do not use firefighting foams. Using firefighting foams on electric vehicles has a higher risk of electrical fire.

8. Towing / transportation / storage

Never transport Semi from the rear. Always transport Semi with front wheels off the ground.

> **WARNING** TO AVOID CAUSING ANY DAMAGE, ENSURE THE TOWING PROCEDURES ARE FOLLOWED. FAILURE TO DO SO CAN LEAD TO SIGNIFICANT DAMAGE AND OVER-HEATING. IN RARE CASES EXTREME OVERHEATING MAY CAUSE THE SURROUNDING COMPONENTS TO IGNITE.

> **WARNING** POSSIBLE BATTERY RE-IGNITION! AFTER A FIRE INCIDENT, STORE OUTSIDE AT A SAFE DISTANCE (150 FT/45 M) FROM OTHER VEHICLES AND STRUCTURES!



> **WARNING** The vehicle is equipped with high voltage components that may be compromised as a result of a collision. Before transporting, it is important to assume these components are energized. Always follow high voltage safety precautions (wearing personal protective equipment, etc.). Failure to do so may result in serious injury.

Preparing Semi for Towing

In situations where there is minimal risk of fire or high voltage exposure and low voltage power is present, Semi can be towed. If Tesla Semi has high voltage power:

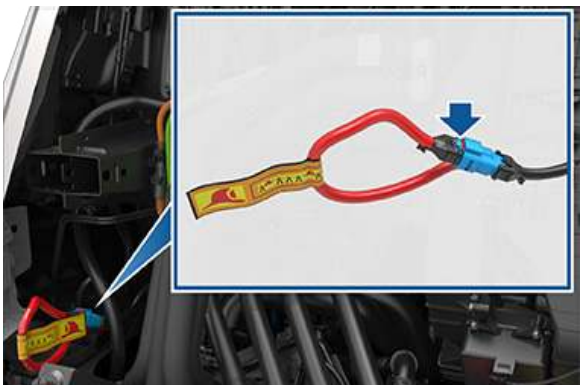
1. Configure the vehicle for towing.
   a. Shift the vehicle into Drive (D) from the gear stalk. Air compressor should turn on.
   b. Shift the vehicle into Neutral (N) by lightly moving the gear stalk.
   c. Press the yellow parking brake button to release the parking brake. The red LEDs bracketing the button will turn off.

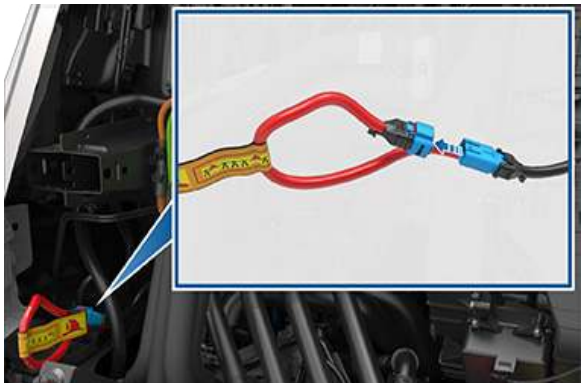


   d. On the left touchscreen, set the Suspension to High.
   e. Re-apply the yellow parking brakes.
   f. Shift vehicle to Accessory state using the gear stalk.
   g. On the right touchscreen, enable **Jack Mode. Touch Controls > Service > Jack Mode**.
   h. On the left touchscreen, open the hood.
   i. On the right touchscreen, power off Semi. **Touch Controls > Safety > Power Off**.

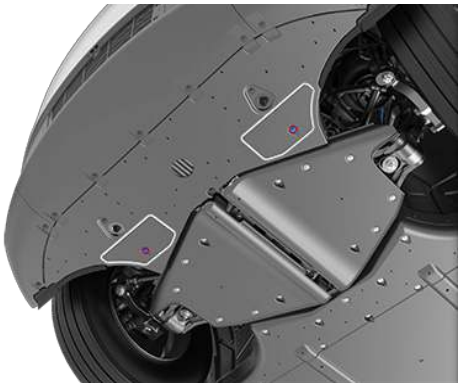
After you've configured Semi for towing or if Semi doesn't have power:

2. Disable high voltage power for Semi.
   a. Disconnect the first responder loop under the hood. You can disconnect the loop by sliding the red tab to the unlocked position, pressing the blue tab, and then pulling off the connector. Double cut the loop if unable to disconnect the loop.
   b. Close the hood.



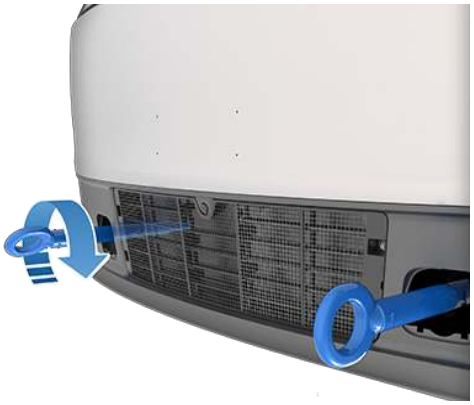


3. Remove the aeroshield subpanels to expose the clevis hook attachments.
   a. Remove the bolts from the subpanels of the aeroshield with a 10mm socket.
   b. Secure the subpanels inside the cabin, or to the exterior back wall of the cabin with ratchet straps.



4. Install the recovery eyes.
   a. Retrieve the recovery eyes from the exterior back wall of Semi.
   b. Remove the rubber caps from the recovery eyes.
   c. Open the recovery eye covers.
   d. Insert both recovery eyes and rotate clockwise until snug.





5. For each rear wheel, remove the half shafts from each rear axle.
   a. For each rear wheel, remove the half shafts from the rear axles.
   b. Position a drip pan against the lower half of the outer wheel.
   c. Remove the axle nuts from each rear wheel.
   d. With both hands, pull the flange towards you to remove the half shaft.

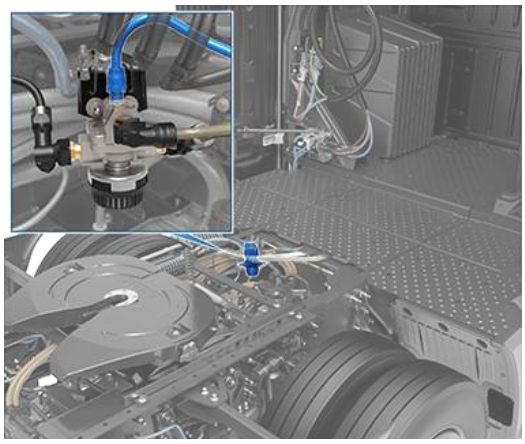


   e. Remove excess oil from the shaft and record the wheel the shaft was removed from.
   f. Stow and secure the shafts in or on the tow vehicle, or on the catwalk of Semi. Store the shafts in garbage bags if possible to avoid spreading oil.
   g. Install the axle covers to prevent leaking of lubricants and damage to wheel bearings.
   h. Remove the drip pan and dispose of any oil according to local regulations.
6. Release or cage the brakes (see "Releasing or Caging the Brakes" in Chapter 8: Towing / Transportation / Storage).
7. Tow Semi (see "Towing Semi" on page 33 in Chapter 8: Towing / Transportation / Storage)

Releasing or Caging the Brakes

The brakes for Semi must be released or caged before towing to allow Semi to roll freely. You can either release the brakes using the quick release valve, or by manually caging the brakes.

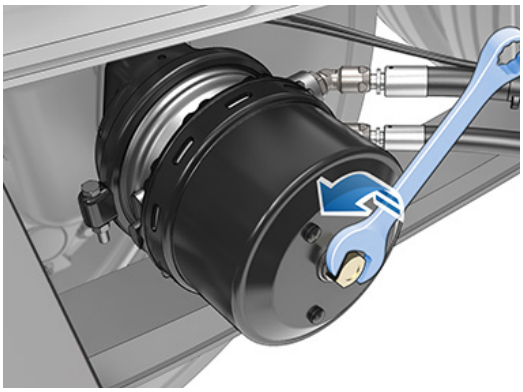
To release the brakes with the quick release valve:
1. Locate the quick release valve on the left side of the vehicle. If a trailer is attached, the valve must be accessed from underneath Semi.
2. Disconnect the air line from the air fitting on the quick release valve supply port.
3. Connect the air supply line of the recovery vehicle to the air fitting. Ensure it is secure.
4. Provide a constant supply of air to Semi from the recovery vehicle. Must be at least 60 psi.
5. Secure Semi to the recovery vehicle. Then remove the wheel chocks.



NOTE: The air supply line is 3/8 inches (1 cm) in diameter.

To cage the brakes:

1. Locate the caging bolt on the brake chamber of the rear wheel.
2. Remove the protective dust cap.
3. Using a 24mm wrench or ratcheting wrench, turn the caging bolt counter-clockwise until it will not turn any further AND confirm that you see a red marking on the caging bolt, indicating the brake spring is mechanically released. Failing to completely release the brakes can cause a fire.
4. Repeat these steps for each rear wheel.

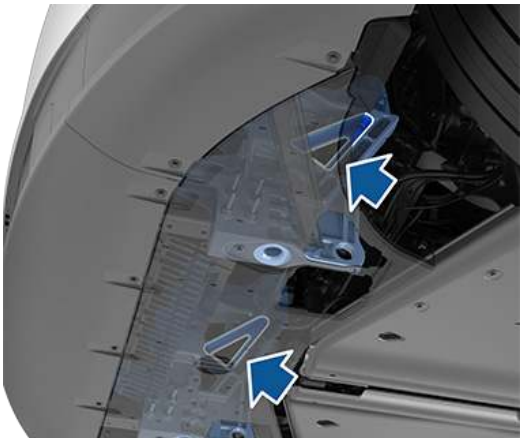


NOTE: Tesla is not responsible for any damage caused by or during transport of the vehicle, including personal property damage.

NOTE: The touchscreen is unresponsive if Semi has no low voltage power. Use an external low voltage power to open the hood and jump start the vehicle's auxiliary low voltage battery.

Towing Semi

1. Attach Semi to the tow vehicle. Front wheels must stay off the ground. Tow bars are recommended, but wheel grids are acceptable.
   a. Align the back of the tow vehicle with Semi.
   b. If the tow truck cannot access the connection points under Semi, connect the tow hooks to the recovery eyes with a double chain or cable setup. Ensure equal weight distribution between recovery eyes. Then, slowly raise the front of Semi 12-18 inches (30-45 cm) from the ground.
   c. Slide the tow vehicle's lower boom under Semi. Align the clevis hooks with the corresponding holes under Semi.
   d. Attach Semi to the clevis hooks.



   f. Secure Semi with emergency chains and straps from the tow vehicle.
   g. Attach any necessary brake or turn lights to the back of Semi, as needed. Semi is not equipped with an electrical connector.
   h. Remove the wheel chocks.

9. Important additional information

This document contains important instructions and warnings that must be followed when handling Semi in an emergency situation.

WARNING Always use appropriate rescue tools and always wear appropriate PPE. Failure to follow these instructions can result in serious injury or death.

WARNING Regardless of the disabling procedure you use, ALWAYS ASSUME THAT ALL HIGH VOLTAGE COMPONENTS ARE ENERGIZED! Cutting, crushing, or touching high voltage components can result in serious injury or death.

WARNING After deactivation, the high voltage circuit requires at least 2 minutes to de-energize.

WARNING The RCM has a backup power supply with a discharge time of approximately 10 seconds. Do not touch the RCM (under the driver's seat) within 10 seconds of pre-tensioner deployment.

WARNING Handling a submerged vehicle without appropriate PPE for water rescue can result in serious injury or death.

WARNING When fire is involved, consider the entire vehicle energized. Always wear full PPE, including a SCBA.

WARNING When cutting the first responder loop, double cut the loop to remove an entire section. This mitigates the risk of the cut wires accidentally reconnecting.

WARNING When using the high voltage shut down methods recommended by this document, high voltage power is isolated to the battery. The high voltage battery is always energized.

WARNING Never transport the Semi with front wheels on the ground. Doing so can lead to significant damage and overheating. In rare cases, extreme overheating may cause the surrounding components to ignite.

Contact Us

First Responders with emergencies, call Tesla Roadside Assistance. Refer to https://www.tesla.com/support/roadside-assistance for the applicable number. Second Responders with emergencies, call Tesla Roadside Assistance. 1-888-TSL-SEMI (1-888-875-7364).

Refer to https://www.tesla.com/firstresponders for the Operator's Manual and first responder information. First responders and training officers with questions, contact firstrespondersafety@tesla.com.

10. Explanation pictograms used

	In some working environments, the Infrared (IR) device is referred to as a Thermal Image Camera (TIC).
	Refers to the hood of a vehicle and follows with detailed procedure for opening the hood both with and without power available.
	Refers to the air suspension of the vehicle and describes the procedure for releasing air.
	Warning
	Flammable
	Explosive
	Corrosive substances present
	Hazardous to human health
	Acute toxicity
	Contains gases under pressure
	Use water to extinguish
	Electricity Warning